

EXPLORATORY ANALYSIS OF CONTACT STRATEGIES FOR ANNUAL ECONOMIC SURVEYS

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coding it forward >



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BACKGROUND

- The Census Bureau utilizes communication strategies to inform and remind potential respondents about the importance of completing surveys.
- However, communication strategies can be resource-intensive and there's limited evidence on how the different components of a strategy impact respondent engagement.



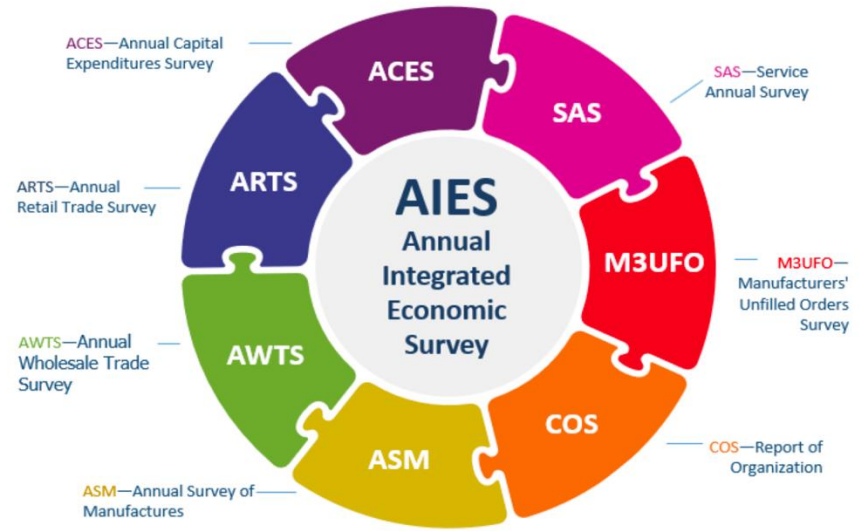
RESEARCH OBJECTIVES

For this project, we aimed to:

- 1. Identify and track trends in communication strategies** over time for several annual economic surveys.
- 2. Better understand the relationship between contact efforts and survey engagement** (authentication code burn rates).
- 3. Provide recommendations on communication strategies best practices** for other surveys at Census, like the Annual Integrated Economic Survey (AIES).

ANNUAL INTEGRATED ECONOMIC SURVEY

- The **Annual Integrated Economic Survey (AIES)**, implemented this year, integrates and replaces seven existing annual business surveys into one survey.
- Survey sent to ~370,000 businesses annually.



DATA SOURCES

Data Sources:

- Collection strategy documentation (survey years 2015-2022)
- Business authentication code usage (survey years 2017-2022)

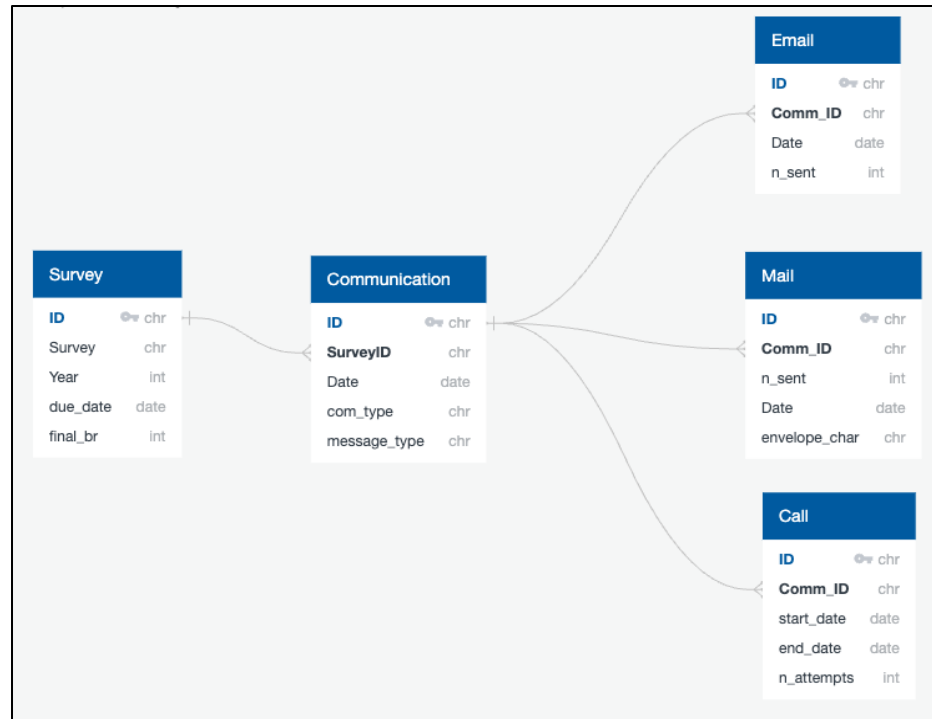
Annual Economic Surveys Included in the Study:

- SAS (Service Annual Survey)
- ARTS (Annual Retail Trade Survey)
- AWTS (Annual Wholesale Trade Survey)
- M3UFO* (Manufacturers' Unfulfilled Orders Survey)
- ACES (Annual Capital Expenditures Survey)
- COS/ASM (Report of Organization/Annual Survey of Manufactures)

*M3UFO was left out of analysis due to lack of historical records.

DATA SOURCES

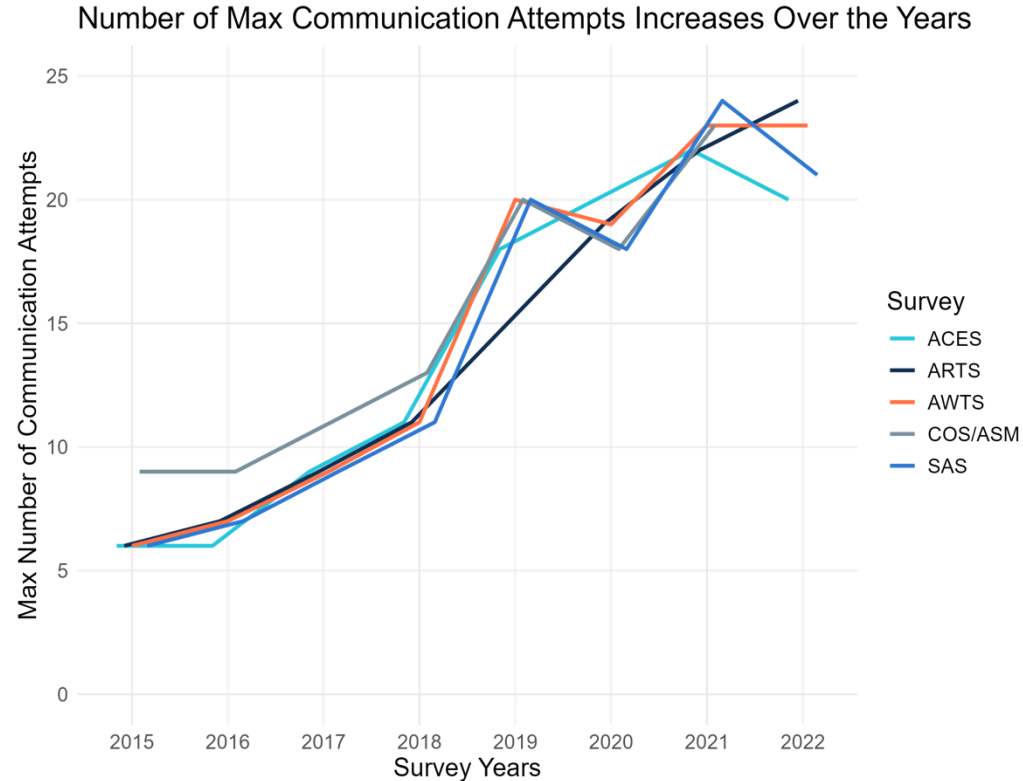
Collection and Outreach Strategy for the <Survey Collection Period & Name> <March 1, 2016 (Date Modified)> <This paragraph should include a high-level summary of the overall collection and outreach strategies and response goals. For example: The Annual Retail Trade Report will continue to follow an all-electronic collection strategy for the 2015 survey year. The only collection instrument used will be Centurion. Respondents will have access to a Question Preview that they can view and/or print. EWD has provided EMD with an overall unit response rate goal of 80%, and rates comparable to the 2014 survey year at each 4 and specific 5 and 6-digit NAICS. This document describes the collection and outreach strategies proposed to achieve these goals.> <Prior Collection Period> Mail Count: <XXX,XXX> Launch Date: <Month Day, Year> Due Date: <Month Day, Year> Soft Closeout Date: <Month Day, Year> Final Closeout Date: <Month Day, Year> Final Unit Response Rate: <XX.X%> Percent Unit Response via Internet: <XX.X%> Percent Unit Response via Mail/Fax/Phone/Analyst: <XX.X%> # Telephone Follow-up (Outgoing) Calls: <XX,XXX> # Telephone Assistance (Incoming) Calls: <XX,XXX> <Current Collection Period> Mail Count: <XXX,XXX> Launch Date: <Month Day, Year> Due Date: <Month Day, Year> Soft Closeout Date: <Month Day, Year> Final Closeout Date: <Month Day, Year> Final Unit Response Rate: TBD Percent Unit Response via Internet: TBD Percent Unit Response via Mail/Fax/Phone/Analyst: TBD # Telephone Follow-up (Outgoing) Calls: TBD # Telephone Assistance (Incoming) Calls: TBD
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Preprocessing included moving data from word documents into a relational database.

COMMUNICATION OVER THE YEARS

- Across surveys, the number of maximum contact attempts have **tripled** between survey years 2015-2022.

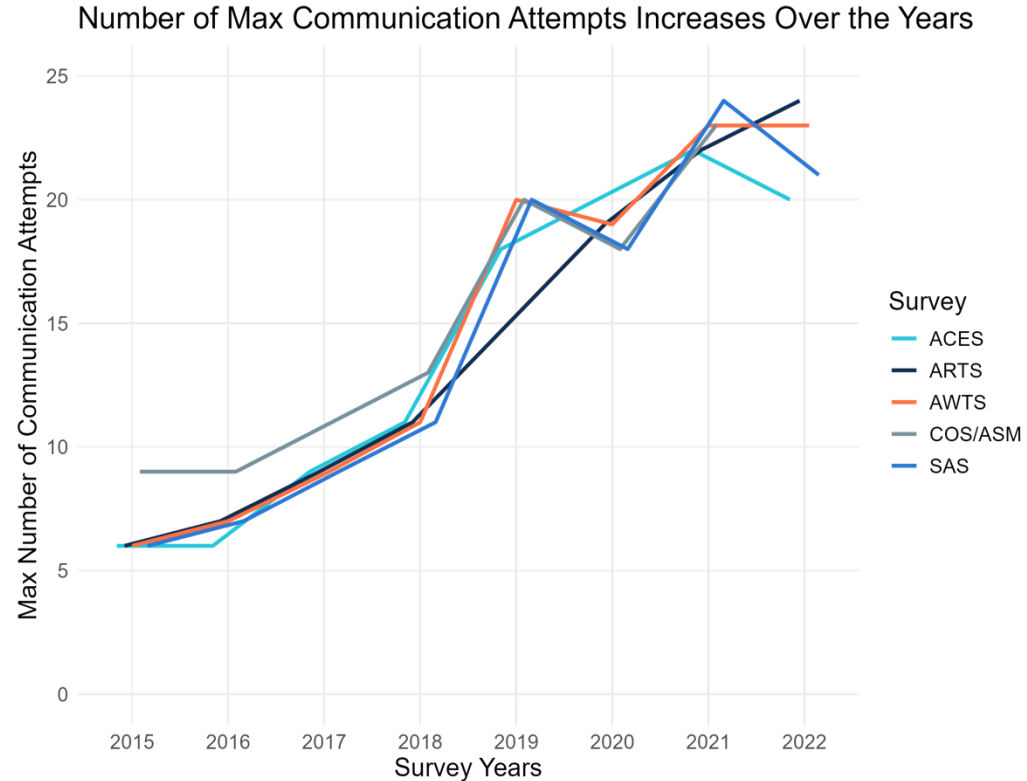


COMMUNICATION OVER THE YEARS

Two notable changes in communication strategies:

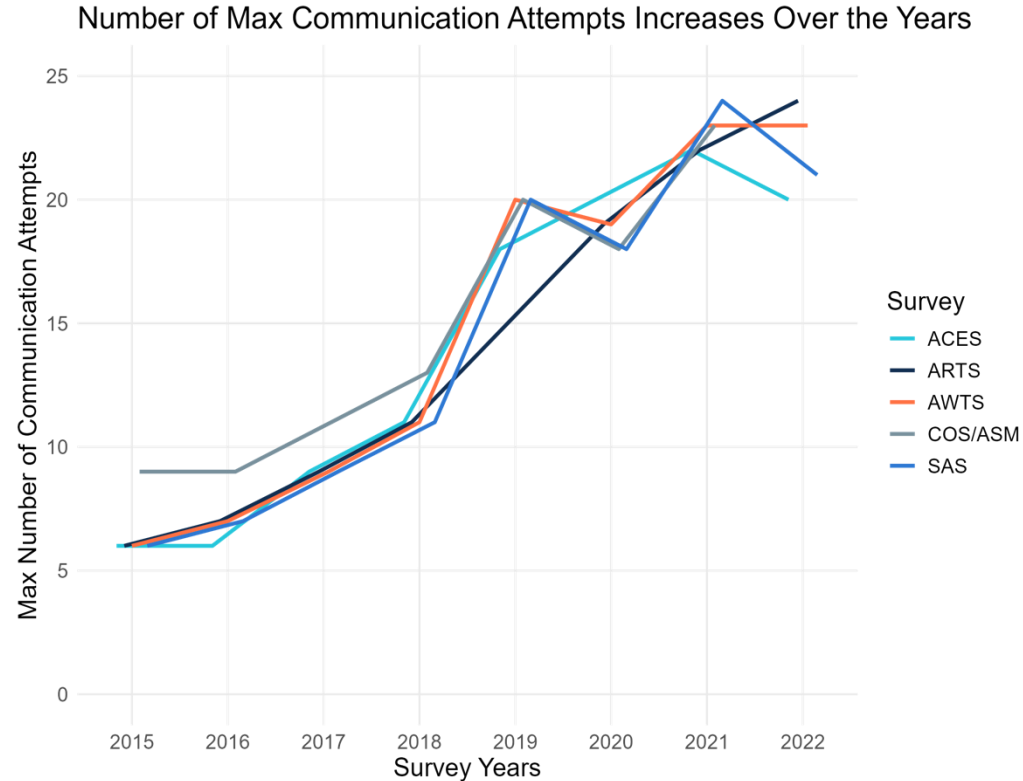
1. Introduction of **email and robocalls** as contact methods

2. Introduction of **new message types** in the communication strategy, like Undeliverable as Addressed mailings.



COMMUNICATION OVER THE YEARS

- Most of the new communication attempts occur **after survey due date.**
- Between survey years 2015-2022, the average number of follow-up messages increased from **4.6** to **13.6**.

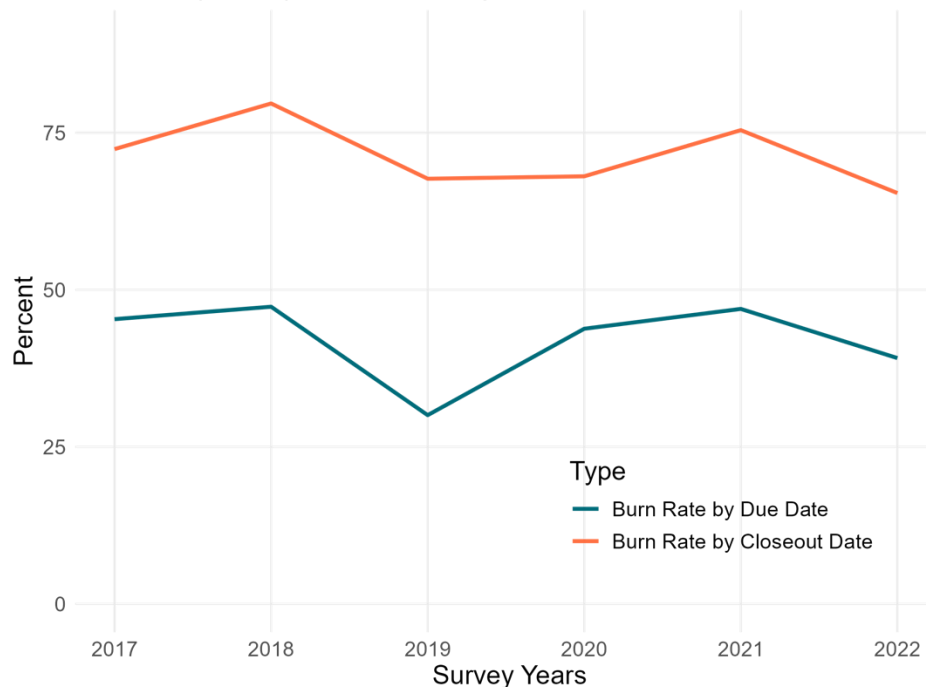


RESPONDENT ENGAGEMENT OVER TIME

- Our outcome of interest is burn rate, the percentage of respondents who use their unique authentication codes to access a survey.
- Calculated burn rate at different points in time to understand engagement across survey period.

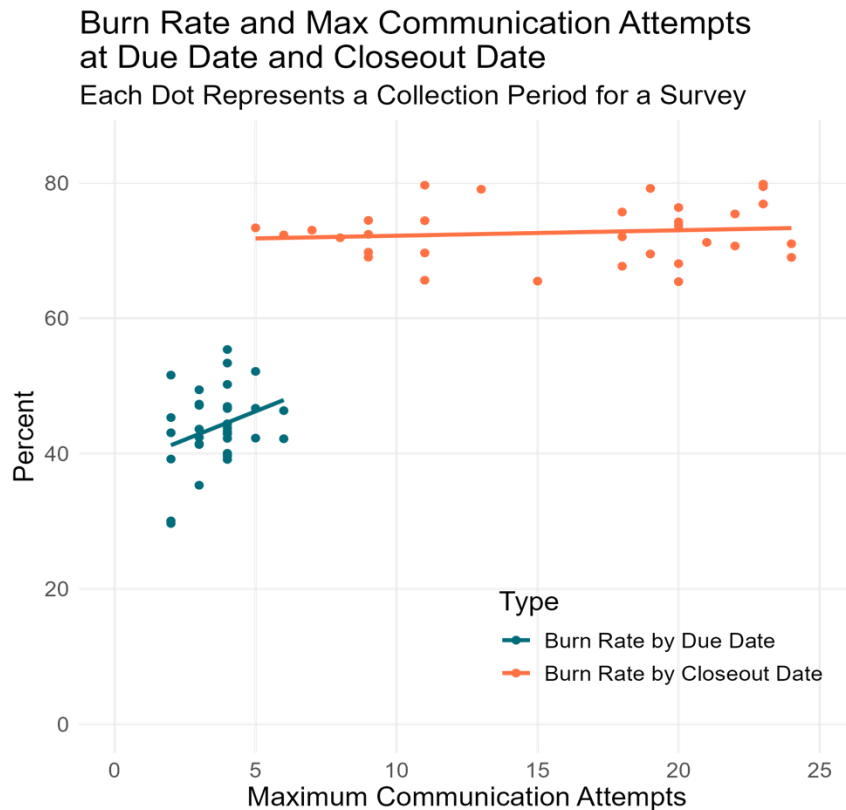
Burn Rate by Due Date and Burn Rate by Closeout
from 2017 to 2022

Annual Capital Expenditures Survey

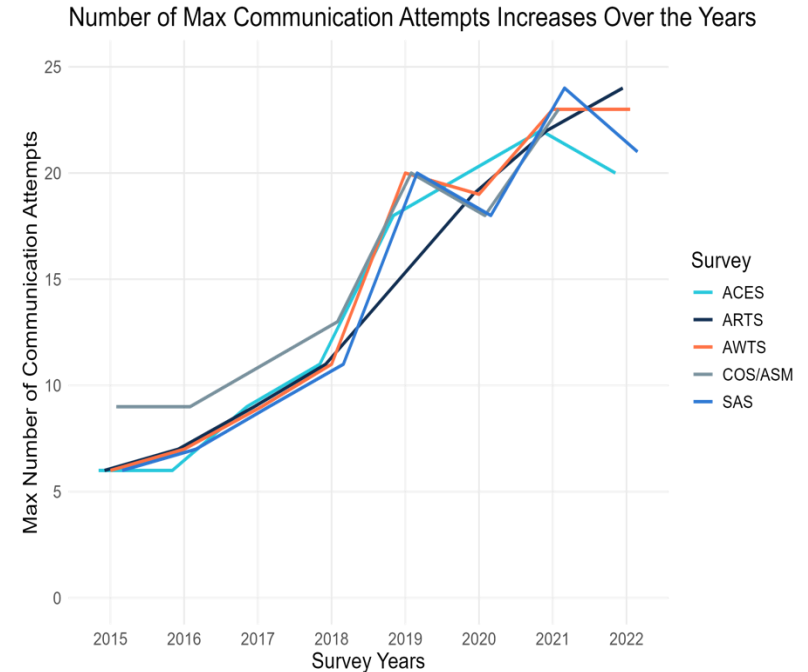
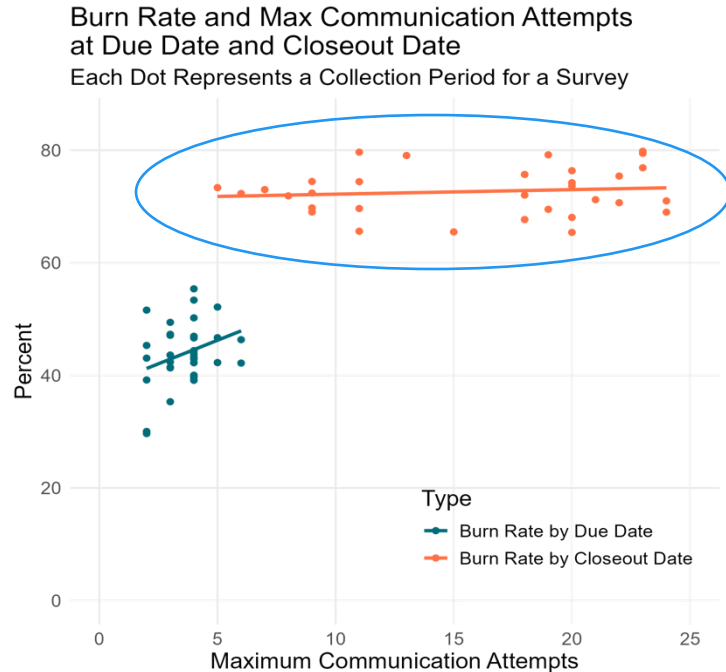


RESPONDENT ENGAGEMENT OVER TIME

- Difficult to discern relationship between all communication attempts and burn rate by survey closeout.
- Subsetting the communication and burn rate data to period between opening and due date shows a more discernible pattern.

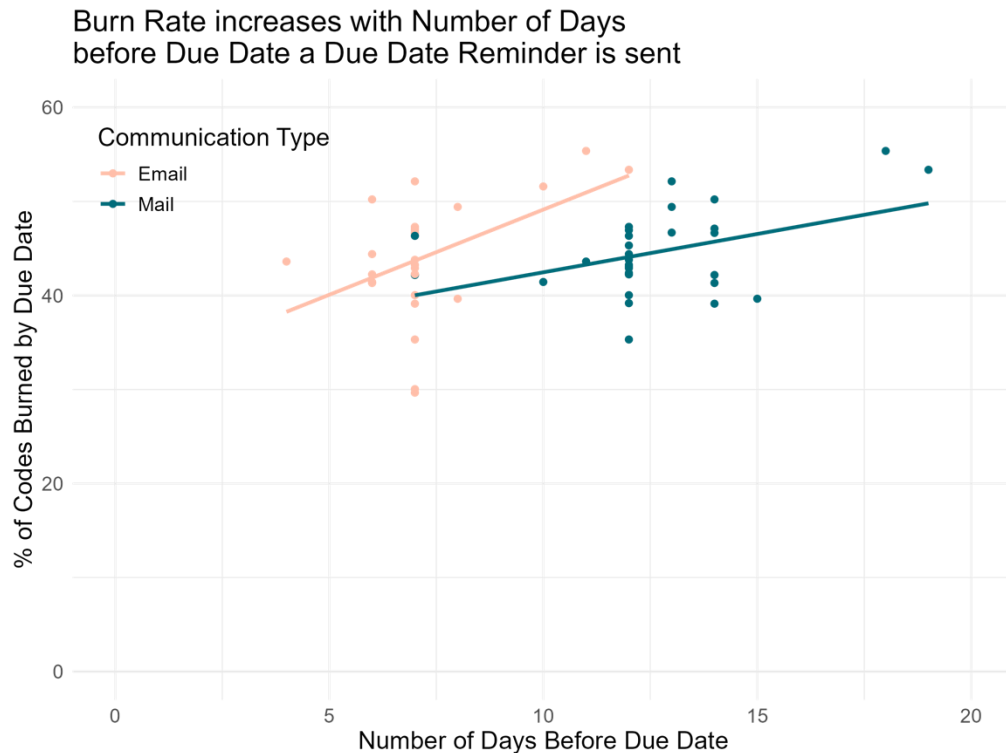


RESPONDENT ENGAGEMENT OVER TIME

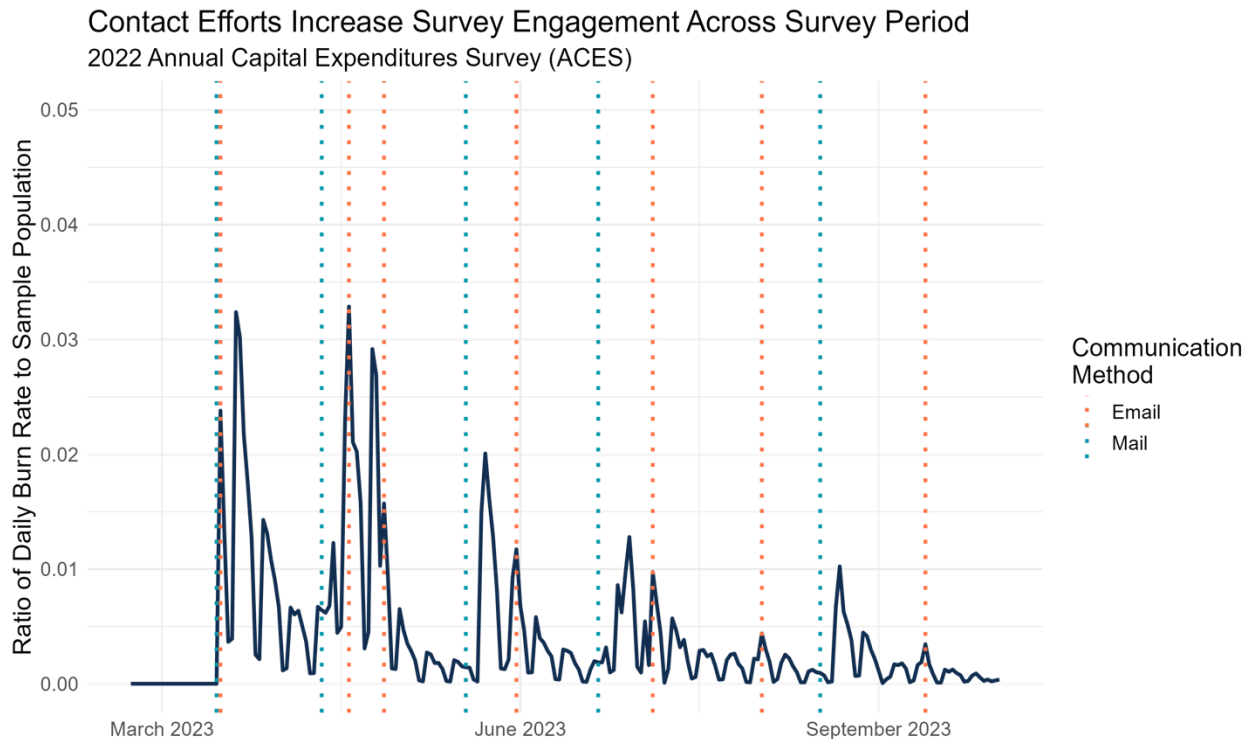


External factors may also be influencing business survey engagement over time.

DUE DATE REMINDER TIMING

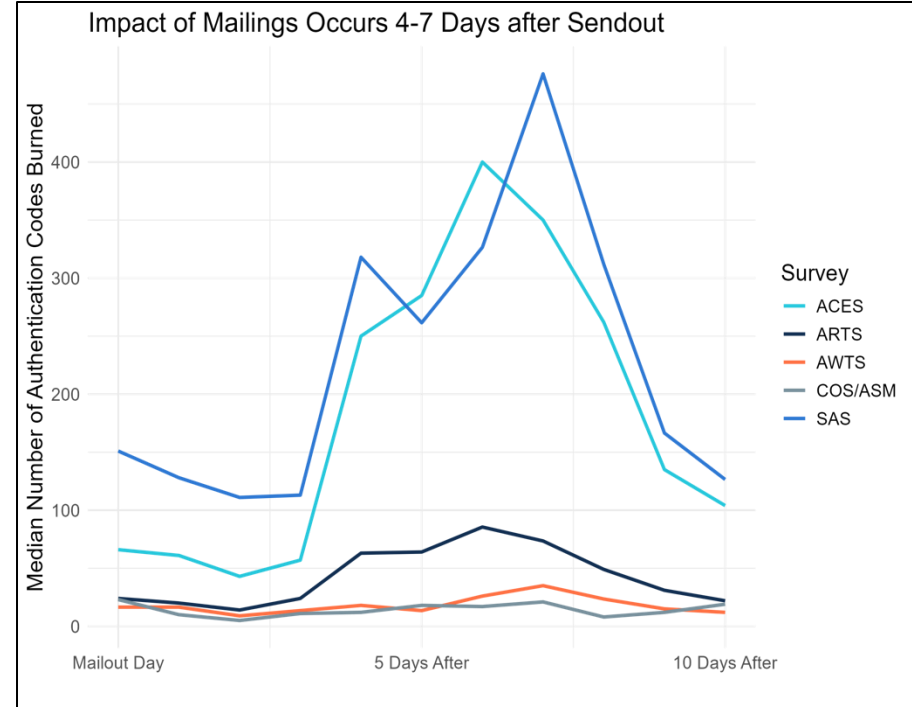
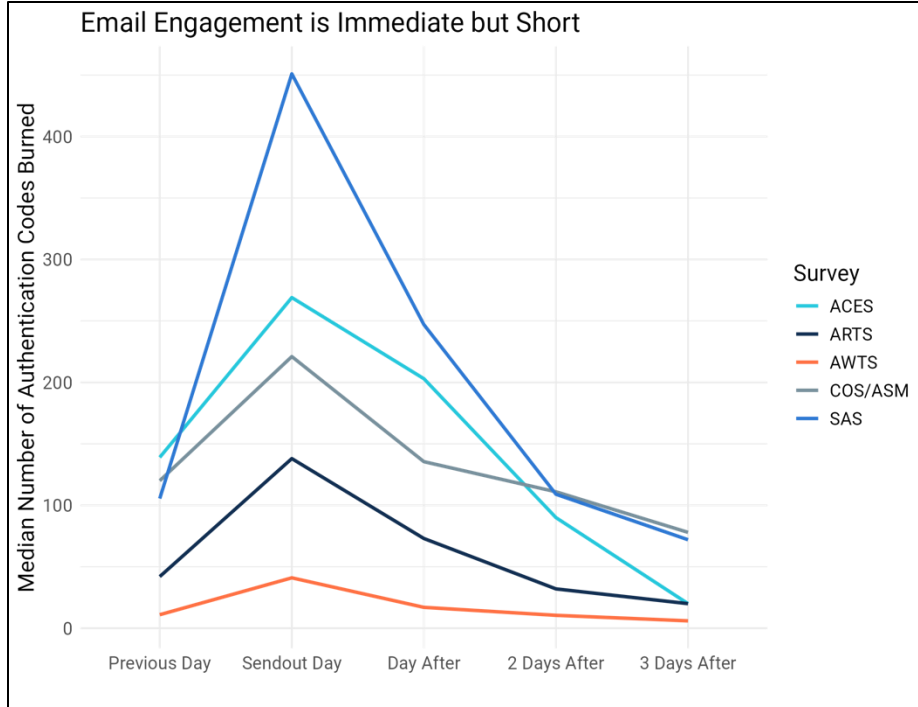


CONTACT EFFORTS AND BURN RATE



Over the course of a collection period, we confirmed that contact efforts correspond to spikes in daily authentication code use.

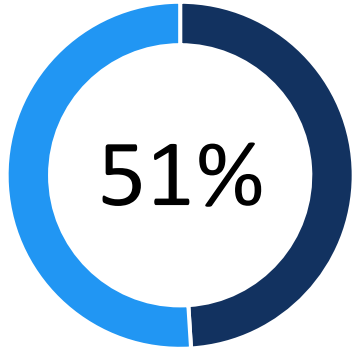
EMAIL VS. MAIL ENGAGEMENT



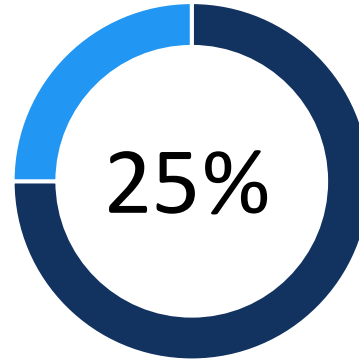
The timing and length of effect of different contact methods varies significantly.

OVERLAPPING COMMUNICATIONS

- When emails and mailings are sent around the same time, it makes it difficult to isolate the impact of one method on authentication code burn rates.
- For **182 mailouts** sent during survey years 2017-2022:



About half of mailouts had at least 1 email sent within 2 weeks of mailing



A quarter of mailouts had at least 1 email sent 3-7 days after mailing (peak activity time)

LIMITATIONS

Due to issues with data accessibility during the summer fellowship, we were unable to:

1. Link respondents with communication they received
2. Include outbound call data in the analysis

As a result, we were only able to investigate the relationship between communication methods and respondent engagement at the survey level (n=28), not at the individual level.

RECOMMENDATIONS

1. Minimize the use of emails in the week following mailouts to better discern the impact that mail specifically has on survey engagement.
2. Maintain historical records for the contact efforts (i.e., lists of businesses that received each communication and date sent) over the statistical period.
3. Store data for the different communication strategies (mail/email/call/robocall) in one parent folder.

CONCLUSION

- Contact efforts notably increase survey engagement but their windows of engagement vary based on the communication method.
- The timing of communications in relation to the survey due date may increase survey engagement and responses; however, more investigation is needed to determine causal effects.
- A next step would be to add additional data sources such as outbound call information to understand the effects that calls specifically have on respondent engagement.



THANK YOU!

If you have questions, I can be reached at
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